

TD1

```
Windows PowerShell
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PS C:\Users\tomda> cd "C:\Users\tomda\Documents\1-ESILV\3-A4\v2\S1\Linux\labs\esilv-marketpulse-g07-til"
PS C:\Users\tomda\Documents\1-ESILV\3-A4\v2\S1\Linux\labs\esilv-marketpulse-g07-til> git branch
  Tom
* main
PS C:\Users\tomda\Documents\1-ESILV\3-A4\v2\S1\Linux\labs\esilv-marketpulse-g07-til> ls

Répertoire : C:\Users\tomda\Documents\1-ESILV\3-A4\v2\S1\Linux\labs\esilv-marketpulse-g07-til

Mode                LastWriteTime         Length Name
----                -
d-----          07/10/2026   18:04             config
d-----          07/10/2026   18:04             data
d-----          07/10/2026   18:04          evidence
d-----          07/10/2026   18:04             labs
d-----          07/10/2026   18:04          readiness
d-----          07/10/2026   18:04             src
-a-----          07/10/2026   18:04           199 .gitignore
-a-----          07/10/2026   18:04          3126 CONTRIBUTING.md
-a-----          07/10/2026   18:04         10747 README.md
-a-----          07/10/2026   18:04           135 requirements.txt
-a-----          07/10/2026   18:27          1510 TEAM.md
-a-----          07/10/2026   18:04          1508 TEAM_TEMPLATE.md

PS C:\Users\tomda\Documents\1-ESILV\3-A4\v2\S1\Linux\labs\esilv-marketpulse-g07-til> git switch Tom
Switched to branch 'Tom'
Your branch is up to date with 'origin/Tom'.
PS C:\Users\tomda\Documents\1-ESILV\3-A4\v2\S1\Linux\labs\esilv-marketpulse-g07-til> pwd

Path
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C:\Users\tomda\Documents\1-ESILV\3-A4\v2\S1\Linux\labs\esilv-marketpulse-g07-til

PS C:\Users\tomda\Documents\1-ESILV\3-A4\v2\S1\Linux\labs\esilv-marketpulse-g07-til> ls

Répertoire : C:\Users\tomda\Documents\1-ESILV\3-A4\v2\S1\Linux\labs\esilv-marketpulse-g07-til

Mode                LastWriteTime         Length Name
----                -
d-----          07/10/2026   18:04             config
d-----          07/10/2026   18:04             data
d-----          07/10/2026   18:04          evidence
d-----          07/10/2026   18:04             labs
d-----          07/10/2026   18:04          readiness
d-----          07/10/2026   18:04             src
-a-----          07/10/2026   18:04           199 .gitignore
```

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Mode                LastWriteTime         Length Name
----                -
d-----          07/10/2026      18:04             config
d-----          07/10/2026      18:04             data
d-----          07/10/2026      18:04             evidence
d-----          07/10/2026      18:04             labs
d-----          07/10/2026      18:04             readiness
d-----          07/10/2026      18:04             src
-a-----          07/10/2026      18:04           199 .gitignore
-a-----          07/10/2026      18:04          3126 CONTRIBUTING.md
-a-----          07/10/2026      18:04          10747 README.md
-a-----          07/10/2026      18:04           135 requirements.txt
-a-----          07/10/2026      18:04          1508 TEAM_TEMPLATE.md

PS C:\Users\tomda\Documents\1-ESILV\3-A4\v2\S1\Linux\labs\esilv-marketpulse-g07-til> ls -la
Get-ChildItem : Impossible de trouver un paramètre correspondant au nom « la ».
Au caractère Ligne:1 : 4
+ ls -la
+ ~~~~~
+ CategoryInfo          : InvalidArgument : (:) [Get-ChildItem], ParameterBindingException
+ FullyQualifiedErrorId : NamedParameterNotFound,Microsoft.PowerShell.Commands.GetChildItemCommand

PS C:\Users\tomda\Documents\1-ESILV\3-A4\v2\S1\Linux\labs\esilv-marketpulse-g07-til> ls -Force

Répertoire : C:\Users\tomda\Documents\1-ESILV\3-A4\v2\S1\Linux\labs\esilv-marketpulse-g07-til

Mode                LastWriteTime         Length Name
----                -
d--h--          07/10/2026      18:41             .git
d-----          07/10/2026      18:04             config
d-----          07/10/2026      18:04             data
d-----          07/10/2026      18:04             evidence
d-----          07/10/2026      18:04             labs
d-----          07/10/2026      18:04             readiness
d-----          07/10/2026      18:04             src
-a-----          07/10/2026      18:04           199 .gitignore
-a-----          07/10/2026      18:04          3126 CONTRIBUTING.md
-a-----          07/10/2026      18:04          10747 README.md
-a-----          07/10/2026      18:04           135 requirements.txt
-a-----          07/10/2026      18:04          1508 TEAM_TEMPLATE.md

PS C:\Users\tomda\Documents\1-ESILV\3-A4\v2\S1\Linux\labs\esilv-marketpulse-g07-til> cd data
PS C:\Users\tomda\Documents\1-ESILV\3-A4\v2\S1\Linux\labs\esilv-marketpulse-g07-til\data> pwd

Path
----
C:\Users\tomda\Documents\1-ESILV\3-A4\v2\S1\Linux\labs\esilv-marketpulse-g07-til\data

```

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Path
----
C:\Users\tomda\Documents\1-ESILV\3-A4\v2\S1\Linux\labs\esilv-marketpulse-g07-til\data

PS C:\Users\tomda\Documents\1-ESILV\3-A4\v2\S1\Linux\labs\esilv-marketpulse-g07-til\data> ls

Répertoire : C:\Users\tomda\Documents\1-ESILV\3-A4\v2\S1\Linux\labs\esilv-marketpulse-g07-til\data

Mode                LastWriteTime         Length Name
----                -
d-----          07/10/2026      18:04         sample

PS C:\Users\tomda\Documents\1-ESILV\3-A4\v2\S1\Linux\labs\esilv-marketpulse-g07-til\data> cd sample
PS C:\Users\tomda\Documents\1-ESILV\3-A4\v2\S1\Linux\labs\esilv-marketpulse-g07-til\data\sample> pwd

Path
----
C:\Users\tomda\Documents\1-ESILV\3-A4\v2\S1\Linux\labs\esilv-marketpulse-g07-til\data\sample

PS C:\Users\tomda\Documents\1-ESILV\3-A4\v2\S1\Linux\labs\esilv-marketpulse-g07-til\data\sample> ls

Répertoire : C:\Users\tomda\Documents\1-ESILV\3-A4\v2\S1\Linux\labs\esilv-marketpulse-g07-til\data\sample

Mode                LastWriteTime         Length Name
----                -
-a-----          07/10/2026      18:04      7625 bloomberg_reference_expected.json
-a-----          07/10/2026      18:04     10281 bloomberg_reference_sample.json
-a-----          07/10/2026      18:04      244 instruments.json
-a-----          07/10/2026      18:04      2266 prices.csv

PS C:\Users\tomda\Documents\1-ESILV\3-A4\v2\S1\Linux\labs\esilv-marketpulse-g07-til\data\sample> c ../../
c : Le terme «c» n'est pas reconnu comme nom d'applet de commande, fonction, fichier de script ou programme
exécutable. Vérifiez l'orthographe du nom, ou si un chemin d'accès existe, vérifiez que le chemin d'accès est correct
et réessayez.
Au caractère Ligne:1 : 1
+ c ../../
+ ~
+ CategoryInfo          : ObjectNotFound: (c:String) [], CommandNotFoundException
+ FullyQualifiedErrorId : CommandNotFoundException

PS C:\Users\tomda\Documents\1-ESILV\3-A4\v2\S1\Linux\labs\esilv-marketpulse-g07-til\data\sample> pwd

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PS C:\Users\tomda\Documents\1-ESILV\3-A4\v2\S1\Linux\labs\esilv-marketpulse-g07-til\data\sample> c ../../
c : Le terme «c» n'est pas reconnu comme nom d'applet de commande, fonction, fichier de script ou programme
exécutable. Vérifiez l'orthographe du nom, ou si un chemin d'accès existe, vérifiez que le chemin d'accès est correct
et réessayez.
Au caractère Ligne:1 : 1
+ c ../../
+ ~
+ CategoryInfo          : ObjectNotFound: (c:String) [], CommandNotFoundException
+ FullyQualifiedErrorId : CommandNotFoundException

PS C:\Users\tomda\Documents\1-ESILV\3-A4\v2\S1\Linux\labs\esilv-marketpulse-g07-til\data\sample> pwd

Path
----
C:\Users\tomda\Documents\1-ESILV\3-A4\v2\S1\Linux\labs\esilv-marketpulse-g07-til\data\sample

PS C:\Users\tomda\Documents\1-ESILV\3-A4\v2\S1\Linux\labs\esilv-marketpulse-g07-til\data\sample> cd ..
PS C:\Users\tomda\Documents\1-ESILV\3-A4\v2\S1\Linux\labs\esilv-marketpulse-g07-til\data> cd ..
PS C:\Users\tomda\Documents\1-ESILV\3-A4\v2\S1\Linux\labs\esilv-marketpulse-g07-til> pwd

Path
----
C:\Users\tomda\Documents\1-ESILV\3-A4\v2\S1\Linux\labs\esilv-marketpulse-g07-til

PS C:\Users\tomda\Documents\1-ESILV\3-A4\v2\S1\Linux\labs\esilv-marketpulse-g07-til> cat .\data\sample\instruments.json
{
  "instrument": {
    "ticker": "AAPL",
    "name": "Apple Inc.",
    "currency": "USD",
    "market": "NASDAQ"
  },
  "benchmark": {
    "ticker": "SP500",
    "name": "S&P 500",
    "currency": "USD",
    "market": "US"
  }
}
PS C:\Users\tomda\Documents\1-ESILV\3-A4\v2\S1\Linux\labs\esilv-marketpulse-g07-til>

PS C:\Users\tomda\Documents\1-ESILV\3-A4\v2\S1\Linux\labs\esilv-marketpulse-g07-til> head data/sample/prices.csv
head : Le terme «head» n'est pas reconnu comme nom d'applet de commande, fonction, fichier de script ou programme
exécutable. Vérifiez l'orthographe du nom, ou si un chemin d'accès existe, vérifiez que le chemin d'accès est correct
et réessayez.
Au caractère Ligne:1 : 1
+ head data/sample/prices.csv
+ ~~~~~
+ CategoryInfo          : ObjectNotFound: (head:String) [], CommandNotFoundException
+ FullyQualifiedErrorId : CommandNotFoundException

PS C:\Users\tomda\Documents\1-ESILV\3-A4\v2\S1\Linux\labs\esilv-marketpulse-g07-til> Get-Content data/sample/prices.csv
date,ticker,open,high,low,close,volume
2026-09-01,AAPL,249.20,251.50,248.00,250.00,38000000
2026-09-01,SP500,6592.00,6615.00,6580.00,6600.00,0
2026-09-02,AAPL,250.80,252.80,249.60,251.30,39500000
2026-09-02,SP500,6607.00,6627.00,6595.00,6612.00,0
2026-09-03,AAPL,249.60,251.30,248.40,249.80,41000000
2026-09-03,SP500,6596.00,6613.00,6584.00,6598.00,0
2026-09-04,AAPL,251.30,253.60,250.10,252.10,42500000
2026-09-04,SP500,6612.00,6635.00,6600.00,6620.00,0
2026-09-08,AAPL,251.90,253.90,250.70,252.40,44000000
2026-09-08,SP500,6623.00,6643.00,6611.00,6628.00,0
2026-09-09,AAPL,254.00,255.70,252.80,254.20,38000000
2026-09-09,SP500,6640.00,6657.00,6628.00,6642.00,0
2026-09-10,AAPL,254.30,256.60,253.10,255.10,39500000
2026-09-10,SP500,6642.00,6665.00,6630.00,6650.00,0
2026-09-11,AAPL,254.20,256.20,253.00,254.70,41000000
2026-09-11,SP500,6642.00,6662.00,6630.00,6647.00,0
2026-09-14,AAPL,256.60,258.30,255.40,256.80,42500000
2026-09-14,SP500,6663.00,6680.00,6651.00,6665.00,0
2026-09-15,AAPL,257.20,259.50,256.00,258.00,44000000
2026-09-15,SP500,6664.00,6687.00,6652.00,6672.00,0
2026-09-16,AAPL,256.70,258.70,255.50,257.20,38000000
2026-09-16,SP500,6663.00,6683.00,6651.00,6668.00,0
2026-09-17,AAPL,259.20,260.90,258.00,259.40,39500000
2026-09-17,SP500,6683.00,6700.00,6671.00,6685.00,0
2026-09-18,AAPL,259.30,261.60,258.10,260.10,41000000
2026-09-18,SP500,6684.00,6707.00,6672.00,6692.00,0
2026-09-21,AAPL,259.10,261.10,257.90,259.60,42500000
2026-09-21,SP500,6683.00,6703.00,6671.00,6688.00,0
2026-09-22,AAPL,261.10,262.80,259.90,261.30,44000000
2026-09-22,SP500,6703.00,6720.00,6691.00,6705.00,0
2026-09-23,AAPL,261.20,263.50,260.00,262.00,38000000
2026-09-23,SP500,6704.00,6727.00,6692.00,6712.00,0
2026-09-24,AAPL,263.00,265.00,261.80,263.50,39500000

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2026-09-29,AAPL,264.50,266.50,263.30,265.00,44000000
2026-09-29,SP500,6729.00,6749.00,6717.00,6734.00,0
2026-09-30,AAPL,266.00,267.70,264.80,266.20,38000000
2026-09-30,SP500,6740.00,6757.00,6728.00,6742.00,0
PS C:\Users\tomda\Documents\1-ESILV\3-A4\v2\S1\Linux\labs\esilv-marketpulse-g07-til> grep AAPL data/sample/prices.csv
grep : Le terme «grep» n'est pas reconnu comme nom d'applet de commande, fonction, fichier de script ou programme
exécutable. Vérifiez l'orthographe du nom, ou si un chemin d'accès existe, vérifiez que le chemin d'accès est correct
et réessayez.
Au caractère Ligne:1 : 1
+ grep AAPL data/sample/prices.csv
+ ~~~~~
+ CategoryInfo          : ObjectNotFound: (grep:String) [], CommandNotFoundException
+ FullyQualifiedErrorId : CommandNotFoundException

PS C:\Users\tomda\Documents\1-ESILV\3-A4\v2\S1\Linux\labs\esilv-marketpulse-g07-til> Select-String -Pattern "AAPL" -Path
"data/sample/price.csv"
Select-String : Impossible de trouver le chemin d'accès «
C:\Users\tomda\Documents\1-ESILV\3-A4\v2\S1\Linux\labs\esilv-marketpulse-g07-til\data\sample\price.csv», car il
n'existe pas.
Au caractère Ligne:1 : 1
+ Select-String -Pattern "AAPL" -Path "data/sample/price.csv"
+ ~~~~~
+ CategoryInfo          : ObjectNotFound: (C:\Users\tomda\...ample\price.csv:String) [Select-String], ItemNotFound
Exception
+ FullyQualifiedErrorId : PathNotFound,Microsoft.PowerShell.Commands.SelectStringCommand

PS C:\Users\tomda\Documents\1-ESILV\3-A4\v2\S1\Linux\labs\esilv-marketpulse-g07-til> Select-String -Pattern "AAPL" -Path
"data/sample/prices.csv"
data\sample\prices.csv:2:2026-09-01,AAPL,249.20,251.50,248.00,250.00,38000000
data\sample\prices.csv:4:2026-09-02,AAPL,250.80,252.80,249.60,251.30,39500000
data\sample\prices.csv:6:2026-09-03,AAPL,249.60,251.30,248.40,249.80,41000000
data\sample\prices.csv:8:2026-09-04,AAPL,251.30,253.60,250.10,252.10,42500000
data\sample\prices.csv:10:2026-09-08,AAPL,251.90,253.90,250.70,252.40,44000000
data\sample\prices.csv:12:2026-09-09,AAPL,254.00,255.70,252.80,254.20,38000000
data\sample\prices.csv:14:2026-09-10,AAPL,254.30,256.60,253.10,255.10,39500000
data\sample\prices.csv:16:2026-09-11,AAPL,254.20,256.20,253.00,254.70,41000000
data\sample\prices.csv:18:2026-09-14,AAPL,256.60,258.30,255.40,256.80,42500000
data\sample\prices.csv:20:2026-09-15,AAPL,257.20,259.50,256.00,258.00,44000000
data\sample\prices.csv:22:2026-09-16,AAPL,256.70,258.70,255.50,257.20,38000000
data\sample\prices.csv:24:2026-09-17,AAPL,259.20,260.90,258.00,259.40,39500000
data\sample\prices.csv:26:2026-09-18,AAPL,259.30,261.60,258.10,260.10,41000000
data\sample\prices.csv:28:2026-09-21,AAPL,259.10,261.10,257.90,259.60,42500000
data\sample\prices.csv:30:2026-09-22,AAPL,261.10,262.80,259.90,261.30,44000000
data\sample\prices.csv:32:2026-09-23,AAPL,261.20,263.50,260.00,262.00,38000000
data\sample\prices.csv:34:2026-09-24,AAPL,263.00,265.00,261.80,263.50,39500000
data\sample\prices.csv:36:2026-09-25,AAPL,262.60,264.30,261.40,262.80,41000000
data\sample\prices.csv:38:2026-09-28,AAPL,263.30,265.60,262.10,264.10,42500000
data\sample\prices.csv:40:2026-09-29,AAPL,264.50,266.50,263.30,265.00,44000000
data\sample\prices.csv:42:2026-09-30,AAPL,266.00,267.70,264.80,266.20,38000000

PS C:\Users\tomda\Documents\1-ESILV\3-A4\v2\S1\Linux\labs\esilv-marketpulse-g07-til> |

PS C:\Users\tomda\Documents\1-ESILV\3-A4\v2\S1\Linux\labs\esilv-marketpulse-g07-til> python --version
Python 3.12.0
PS C:\Users\tomda\Documents\1-ESILV\3-A4\v2\S1\Linux\labs\esilv-marketpulse-g07-til> git --version
git version 2.51.1.windows.1
PS C:\Users\tomda\Documents\1-ESILV\3-A4\v2\S1\Linux\labs\esilv-marketpulse-g07-til> git --version
git version 2.51.1.windows.1
PS C:\Users\tomda\Documents\1-ESILV\3-A4\v2\S1\Linux\labs\esilv-marketpulse-g07-til> python src/main.py
=== MarketPulse ===

Instrument
AAPL - Apple Inc.
Last price: 266.20 USD

Benchmark
SP500 - S&P 500
Last level: 6742.00

Period: 1 month
Interval: Daily

Observations
AAPL: 21
SP500: 21
PS C:\Users\tomda\Documents\1-ESILV\3-A4\v2\S1\Linux\labs\esilv-marketpulse-g07-til>

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